

Can We Save The Public Internet?

What is the problem with the public Internet?

As a quick scan of the recent SIGCOMM program will reveal, much research today focuses on the technology for individual networks – such as programmable routers, verification techniques, and load balancing algorithms – and on protocol-specific challenges – such as BGP configuration, video adaptation, and congestion control. In contrast, there is little current research on the conceptual foundations of the Internet itself. That is, we as a community largely accept as given how these individual networks coordinate to form a unified substrate on which applications can be deployed. This acceptance is natural because the Internet’s performance, availability, and capacity have been consistently improving over the past decade, often due to innovations in the Internet’s individual networks and protocols derived from our community’s research. However, as this essay describes, this progress has come at a steep but hidden price that imperils the future of what we call the “public Internet.”

To understand why this seemingly beneficial progress is so problematic, recall that the Internet was originally designed to provide end-to-end packet delivery by interconnecting independent networks, which gave rise to the term “Internet.” By interconnection, we mean that an individual network could join the Internet by merely connecting to another network that was already part of the Internet; interdomain routing ensured that packets would be forwarded to the appropriate network. This form of interconnection allowed the Internet’s coverage to expand organically but relied on the presence of transit ISPs that were willing, for a fee, to carry traffic that neither originated nor terminated in their domain. These transit ISPs allowed the Internet to achieve global coverage while not requiring any individual network to have a global scope. We will call this delivery-via-transit-ISPs paradigm the “public Internet” because: (i) while the individual ISPs were typically commercial, this paradigm offered a way for all networks, no matter how small, to join the Internet as long as they were willing to pay the access cost, and (ii) this infrastructure provided a substrate on which anyone could deploy applications, again as long as they were willing to pay their access fee.

However, the Internet of today is quite different from this original vision. Rather than providing pure end-to-end packet delivery, today’s Internet provides many “enhancements,” most notably support for content delivery (e.g., via CDNs) but also various security functions (such as DoS protection, in-network firewalls, and zero-trust network access). These enhancements are typically deployed at network edges on clusters of general-purpose servers and interpose on traffic going between clients and backend datacenters. Thus, the idealized notion of the Internet providing pure end-to-end packet delivery has been irretrievably abandoned (as noted in the seminal 2001 paper by Blumenthal and Clark).

Note that these enhancements do not interconnect in the way that the public Internet does; there is no global routing between these clusters that ensures that the appropriate enhancements are applied before packets are delivered. Instead, each enhancing infrastructure acts independently (i.e., each CDN handles routing between their own clusters, but CDNs do not coordinate with each other to make sure content requests are directed to the appropriate cache). Thus, global coverage for such enhancements requires either that the enhancement provider be global or at least very large so a content provider can contract with a small set of them. The incentives have long pointed strongly towards the deployment of a set of large and parallel enhancing infrastructures, and this is what the early set of CDNs was.

The nature of these enhancing deployments has changed over time. In fact, CDNs were deployed soon after the Internet went commercial and were instrumental in allowing the Internet to handle the rapid growth of content demands. These early CDNs used the public Internet to connect their various edge-based clusters; as such, they relied on the public Internet rather than trying to displace it.

In the past decade, a new generation of enhancing infrastructures has emerged – from various hyperscaled cloud and content providers such as Amazon, Azure, Google, and Meta – that integrates a set of enhancing clusters with their own private networks; these private cluster-connecting networks are distinct from their backbone networks connecting their datacenters. These enhancing private networks do not offer transit; instead, they only carry traffic to and from their own network. They, too, must be large-scale and deployed in parallel.

Such enhancing private networks are so prevalent that the vast majority of traffic (estimated at well over 70%) does not utilize a transit network but is either handled by a cluster within the originating network or goes directly between an access ISP and one of these new enhancing private networks. As a result, the role of the public Internet is rapidly diminishing.

We are thus on a trajectory where in the relatively near future almost all traffic will travel directly between an access network (where clients attach) and one of the few large-scale enhancing private networks. Such an Internet has two significant drawbacks.

First, this result would greatly exacerbate the digital divide. The reach of these enhancing private networks is even

more limited than that of high-bandwidth connections and is much harder to extend because, as mentioned previously, these enhancements do not interconnect; in fact, each of these enhancing private networks would need to extend into every region in order to provide universal coverage. This is an alarming and unacceptable prospect.

Second, the necessity of building large-scale infrastructure poses a significant barrier to entry into the market. Only the very largest entities will be able to deploy such enhancing private networks, and only the services they offer (or support on their cloud computing backend) will garner the advantages of their enhancements. The resulting Internet would no longer provide a level playing field for applications, which betrays the early promise of the Internet.

What is the Grand Challenge? The grand challenge before us is to combine the interconnection paradigm of the public Internet with the enhanced functionality of the enhancing private networks. The result would be an enhanced public Internet that would offer an advanced set of enhancements that are available to all applications and that evolve over time to meet new needs. In addition, the enhanced public Internet could extend its coverage through its ability to interconnect networks, so it need not have global infrastructures to achieve global coverage. This combination of features would be of huge benefit to society and foster even faster innovation of Internet-based services.

Where Is the research?

Meeting this challenge requires a deep and broad set of research efforts. To be clear, however, this is not a clean-slate research agenda that starts from scratch. Building an enhanced public Internet requires a new conceptual framework for how to think about an enhanced public Internet, but the actual designs should be incrementally deployable and backwards-compatible, not requiring any change to the basic packet-delivery functions of the current Internet. Below, we identify six areas where research is crucially needed.

First, there is a basic architectural challenge: How does one combine enhancements with interconnection to create an evolvable and enhanced public Internet? This challenge is not met by merely requiring standards for the enhancements but requires (i) an understanding of how these enhancing clusters coordinate with each other so that the enhancements are applied at the appropriate network edges, (ii) an ability to allow new enhancement functionality while retaining uniformity, and (iii) an understanding of how money flows so that there are incentives for deployment. All three of these are significant and currently unaddressed challenges.

Second, an enhanced public Internet requires the interposition of enhancements between clients and servers. This interposition requires a rethinking of traditional protocols, particularly when the enhancement provider is different from the content provider (so there are three trust domains in an exchange: the client, the backend server, and the enhancing edge). We need research to understand how to design protocols that cleanly and efficiently support such interposition.

Third, such an enhanced public Internet poses both challenges and opportunities for security. The challenge is to modify basic security and privacy protocols to adapt to the kinds of interpositions the enhancements require. The opportunity is that the general-purpose clusters can be equipped with secure enclaves, which will enable new approaches to mixnets, remote attestation, and other security challenges.

Fourth, while the enhancements are currently performed on general-purpose clusters, eventually, hardware support will be needed to achieve higher efficiency. This will require a better understanding of how to transition software implementations to a variety of accelerated hardware platforms while retaining compatibility with pure software implementations. This is a very different challenge than writing code for a specific programming model such as P4 or eBPF. Here, the challenge is now to structure traditional software so that individual components can be easily and compatibly accelerated by a variety of accelerator technologies.

Fifth, rather than waiting for the commercial world to show us how the network can enhance application performance (which has been the case with our current set of enhancements), the research community should take the lead in exploring new use cases. These could range from supporting gaming via new multiparty delivery services, to enhancing federating learning through in-network aggregation, to leveraging secure enclaves as described above. The fact that these enhancements would be standardized means that basic research could have an influence on their trajectory, as opposed to today's set of proprietary enhancement designs.

Sixth, the last challenge is enabling the deployment of the enhanced public Internet. This is far more complicated than deploying a single piece of standalone technology, where academic prototypes can lead to startups that disrupt an industry through better performance on a particular isolated task. In contrast, the adoption of the enhanced public Internet would be an example of a Technology Ecosystem Transformation (TET) led by the research community, which requires a far more delicate interplay between academic prototypes, research testbeds, early adoption by fringe elements of the ecosystem, fully commercial adoption by some major players in the ecosystem, and then finally a more complete adoption as the enhanced public Internet takes over as the dominant Internet paradigm. Yes, this series of

events may sound far-fetched, but this is exactly what happened with the Internet itself, and it is time to repeat that research success, led by NSF.